

Software Architecture & System Planning Manual

A structured engineering blueprint for CTOs, VPs of Engineering, and IT Leaders planning high-scale web platforms, mobile apps, and spatial IT infrastructure.

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SECTION 01

Microservices vs. Monolith Architecture Engine

Choosing between a monolithic architecture and a microservice-oriented layout depends on operational complexity, team velocity, and concurrency demands.

CRITERIA	MODULAR MONOLITH	MICROSERVICES MESH
Team Size	< 15 Engineers	20+ Multi-Discipline Teams
Concurrency	< 10,000 req/sec	100,000+ req/sec
Deployment	Single Pipeline	Independent CI/CD per service
Data Consistency	ACID / Relational DB	Eventual Consistency / Kafka

SECTION 02

Cloud Infrastructure & Performance Benchmarks

Estimated monthly AWS infrastructure costs based on modern production workloads (React 19 Frontend + Node.js Microservices).

STARTER SCALE Target Workload: 10K MAU AWS ECS Fargate, Single Aurora PostgreSQL, CloudFront CDN, Redis.	\$350 - \$800 / mo
ENTERPRISE MID-SCALE Target Workload: 250K MAU EKS Kubernetes Mesh, Multi-AZ Aurora, ElastiCache Cluster, WAF.	\$2,500 - \$6,000 / mo
GLOBAL HYPER-SCALE Target Workload: 5M+ MAU Multi-Region Kubernetes, DynamoDB Global Tables, Cloudflare Enterprise.	\$18,000+ / mo

SECTION 03

Enterprise Security & Audit Compliance Checklist

[] ISO 27001 & SOC 2 Type II

Enforce AES-256 encryption at rest, TLS 1.3 in transit, automated vulnerability scanning in CI/CD pipelines.

[] Identity & Access Management

OAuth 2.0 + OIDC authentication, RBAC with least privilege model, mandatory Multi-Factor Authentication (MFA).

[] HIPAA & GDPR Data Governance

Anonymized data logging, customer consent management, right-to-be-forgotten automated purges.

SECTION 04

Vendor RFP & Selection Scorecard Matrix

Use this weighted scoring system when evaluating software engineering partners:

Technical Architecture Depth (Senior Engineering Ratio)	Weight: 30%	Target: 8/10
Security & Compliance Maturity (SOC 2, ISO Compliance)	Weight: 25%	Target: 9/10
Delivery Speed & Agile Sprint Discipline	Weight: 25%	Target: 8/10
Post-Launch SLA & 24/7 Infrastructure Support	Weight: 20%	Target: 8.5/10

Need Expert Architecture Guidance?

Riyadvi Software Technologies partners with enterprises globally to architect, build, and scale world-class web applications, mobile platforms, and AI systems.

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